

MELSOM, NorwayWMO: **014810**Lat: **59.230N**Lon: **10.348E**Elev: **26**StdP: **101.01**Time Zone: **1.00 (EUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.4	-17.5	0.8	-14.1	-15.0	1.0	-11.4	7.5	5.5	6.7	5.3	1.1	0	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	26.2	17.4	24.2	16.7	22.3	15.9	18.7	23.9	17.8	22.3	16.9	20.7	1.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	20.2	16.0	11.4	19.3	15.2	10.8	18.5	53.1	24.2	50.2	22.5	47.5	20.7	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.1	5.2	4.5	-17.4	28.4	3.2	2.2	-19.7	29.9	-21.6	31.2	-23.4	32.5	-25.7	34.1		
			-17.6	20.0	3.1	1.2	-19.8	20.8	-21.7	21.5	-23.4	22.2	-25.7	23.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.9	-1.6	1.5	6.2	11.2	15.0	17.3	16.2	12.7	7.6	2.9	-0.6
	DBStd	7.79	5.00	4.69	4.27	3.07	3.56	2.52	2.53	2.08	2.57	3.19	3.82	5.40
	HDD10.0	1739	369	324	266	118	26	0	0	0	8	87	213	328
	HDD18.3	4069	628	558	523	363	222	104	51	71	170	332	462	586
	CDD10.0	744	0	0	1	6	63	151	228	193	87	13	1	0
	CDD18.3	32	0	0	0	0	2	5	20	6	0	0	0	0
	CDH23.3	266	0	0	0	0	21	64	149	32	0	0	0	0
	CDH26.7	43	0	0	0	0	1	8	32	2	0	0	0	0
(14) Wind	WSAvg	2.1	2.3	2.2	2.2	2.1	2.1	1.9	1.7	1.7	1.9	2.1	2.4	2.3
(15) Precipitation	PrecAvg	974	85	60	48	53	63	68	79	118	92	112	111	85
	PrecMax	1364	189	185	100	115	153	131	175	227	283	312	212	182
	PrecMin	726	14	7	1	14	2	5	12	33	38	7	18	27
	PrecStd	178	46	42	30	27	38	31	35	49	62	70	49	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	10.1	15.6	18.7	25.9	27.7	29.9	26.3	21.1	16.1	12.4	10.2
		MCWB	6.3	5.6	8.9	10.6	15.8	16.6	18.9	18.6	15.2	12.1	10.4	8.4
	2%	DB	7.2	7.3	11.9	15.8	22.7	25.3	27.1	23.8	19.5	14.6	11.3	8.9
		MCWB	5.6	4.1	6.7	9.1	14.5	16.2	18.3	17.1	14.4	11.9	9.5	7.5
	5%	DB	6.1	5.6	8.9	13.8	20.1	22.5	24.8	22.0	18.3	13.6	10.1	7.9
		MCWB	4.6	3.6	4.6	8.3	13.3	15.0	17.3	16.3	13.8	11.5	8.7	6.5
	10%	DB	4.9	4.5	7.0	12.0	17.7	20.5	22.8	20.7	17.1	12.7	8.8	6.6
		MCWB	3.3	3.1	3.8	7.4	12.1	13.9	16.7	15.8	13.6	10.8	7.4	5.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	6.5	9.3	11.9	17.3	18.6	20.1	19.5	17.0	13.9	11.6	9.0
		MCDB	8.0	8.7	15.1	15.8	23.2	24.2	26.2	23.4	18.8	14.9	12.1	9.9
	2%	WB	6.0	4.9	6.9	10.2	15.6	17.1	19.1	18.2	15.9	12.9	9.7	7.8
		MCDB	7.0	6.3	11.1	14.4	20.6	23.3	25.1	21.9	17.9	14.0	10.5	8.7
	5%	WB	4.8	4.1	5.5	9.1	14.3	16.0	18.3	17.2	15.1	12.1	8.6	6.7
		MCDB	5.8	5.2	7.9	13.1	18.9	20.9	23.4	20.5	17.1	13.3	9.6	7.7
	10%	WB	3.5	3.3	4.5	8.0	12.9	15.0	17.3	16.5	14.3	11.1	7.4	5.3
		MCDB	4.7	4.4	6.5	11.3	16.6	19.0	21.5	19.4	16.3	12.5	8.6	6.4
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	7.7	8.3	8.7	8.9	8.5	8.2	7.7	6.2	5.1	5.2
		MCDBR	5.7	7.2	10.6	11.3	12.4	12.5	11.9	10.2	9.4	7.1	6.4	5.5
		MCWBR	5.3	5.2	6.5	6.2	5.8	5.8	5.0	5.0	5.6	5.2	5.8	5.2
		MCDBR	5.4	5.7	9.0	10.1	10.6	10.3	10.4	8.4	7.4	5.8	5.8	5.4
	5% WB	MCWBR	5.3	4.5	6.0	5.9	5.4	5.6	4.9	4.7	5.2	4.9	5.7	5.4
(40) Clear-Sky Solar Irradiance	taub		0.280	0.317	0.337	0.377	0.371	0.358	0.380	0.376	0.346	0.333	0.298	0.256
	taud		2.233	2.309	2.355	2.312	2.410	2.468	2.427	2.434	2.505	2.437	2.293	2.158
	Ebn at Noon		527	697	800	824	859	877	845	820	791	677	501	412
	Edh at Noon		43	68	87	107	104	100	102	95	75	60	41	32
(44) All-Sky Solar Radiation	RadAvg		0.31	0.90	2.41	3.86	5.09	5.81	5.20	4.09	2.71	1.21	0.42	0.21
	RadStd		0.06	0.17	0.24	0.53	0.51	0.39	0.61	0.28	0.30	0.21	0.10	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page